JCE Infrastructure Pty Ltd

CIVIL · HYDRAULICS · INFRASTRUCTURE ENGINEERING

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FEE PROPOSAL AND METHODOLOGY

Infrastructure Asset Condition Assessments and Valuation

Murray River Council · Tender MRC-2645 (VendorPanel VP527277) · Schedule 6 methodology and basis of the Schedule 1 price

Proposal No	JCE-FP-VP527277	**Client**	Murray River Council
Date	12/10/2026	**Contact**	Peter Beaumont, Procurement Lead (via VendorPanel)
Revision	H (final for review and signature; JCE vehicle and drone capture, GIS)	**Project**	MRC-2645 Transport Infrastructure Assessment and Valuation 2026
Prepared and reviewed by	Jayden Callen, Director	**Site**	Murray River LGA: Barham, Mathoura, Moama and Moulamein zones
Tender closes	2:00 pm Friday 13/11/2026 (VendorPanel)	**Lump sum (incl. GST)**	$ 1,270,973.00

Cover letter

Mr Peter Beaumont, Procurement Lead
Murray River Council, 52 Perricoota Road, Moama NSW 2731
By VendorPanel (VP527277)

Dear Mr Beaumont,

Tender MRC-2645: Infrastructure Asset Condition Assessments

JCE Infrastructure Pty Ltd (JCE) tenders to inspect, condition rate and value all of Council's transport and bridge infrastructure as at 30/06/2027, and then to deliver the four rolling zone inspections and valuations from 2028 to 2031. JCE offers one supplier, one lump sum and one point of contact, as the Conditions of Tender require.

I lead every JCE commission personally. My recent roles have been in design management: coordinating flood, electrical, hydraulic and structural subconsultants into single packages at Langtree Consulting Engineers (2025 to 2026), and managing a multidisciplinary design team and external subconsultants on Defence projects for Lendlease and CPB Contractors at SCP Consulting (2023 to 2025). On MRC-2645 I will run JCE's own road and drone capture, the bridge inspection partner and the valuer to one program, one rating guide and one quality process, and I will be accountable to Council for all of it. I have also worked for a regional council in Far North Queensland, where I shortlisted consultants for a council tender, so I know what Council needs from its consultant after award: clear hold points, no surprises on scope or fee, and records Council can audit.

JCE captures about 3,000 km of road itself with a vehicle-mounted geo-referenced camera rig, and drone imagery of bridges and approaches, culverts, stormwater outlets, kerb and footpath hotspots and assets with poor vehicle access. JCE rates every asset class except the bridges, builds the GIS and asset register, and carries the methodology, Council liaison, data quality, remaining life, defect and safety assessment, the 10-year programs and the reporting. The bridge partner carries out Level 2 inspections of 127 bridges, major culverts and footbridges, and a qualified valuer signs the valuation.

The lump sum is $ 1,270,973.00 including GST for all five valuation years, with Level 3/4 bridge inspections priced separately as Council's option. Our offer stays open for 60 days after the closing time.

Yours faithfully,

[signature]
Jayden Callen
Director, JCE Infrastructure Pty Ltd
0477 207 027 · jayden@jceinfrastructure.com.au

1.0 Project understanding

Council requires a comprehensive review, inspection, condition assessment and comprehensive valuation of all transport and bridge infrastructure across the Murray River LGA, followed by a four-year rolling program of zone inspections and annual valuations. Every asset is inspected and recorded, rated on Council's 1 (Excellent) to 5 (Very Poor) scale, given a remaining useful life, and valued at fair value under AASB 13, AASB 116 and the Local Government Code of Accounting Practice and Financial Reporting. Assets below Council's capitalisation thresholds (Appendix 3) are still recorded, with condition and remaining life, and flagged "not to be valued".

Council's Appendix 1 sets out the network (quantities to be verified by inspection):

Asset class	Quantity (Appendix 1)	Inspection and rating	Components (Appendix 2)
Bridges and major culverts	62 concrete, 39 composite, 1 timber, 11 major culverts (113)	Level 2 inspection	Substructure, superstructure, deck, abutments, miscellaneous
Footbridges	9 steel, 5 timber (14)	Level 2 inspection, handrails	As for bridges
Sealed roads	1,110 km (urban 133, regional 416, local 561)	Images every 10 m; urban block to block, rural 2 km segments	Surface, pavement, formation
Unsealed roads	1,890 km	Images every 10 m	Wearing course, formation
Kerb and gutter	176 km	Per segment	Single component
Footpaths and cycleways	121 km	Per segment	Single component
Roundabouts, traffic islands, medians	21; 199; 29	Per asset	Single component
Carparks	63	Per asset and component	Sealed or unsealed components
Rural culverts	About 1,555 (to be confirmed)	Per asset	Single component
Guardrail	About 500 items (to be confirmed)	Condition and safety assessment	Single component
Bus shelters	About 35 (to be confirmed)	Per asset	Single component

The network spans four zones: Barham (809 km of road), Mathoura (864 km), Moama (474 km) and Moulamein (842 km). Barham Road, Wakool Road and Moulamein Road are collected whole in one zone, and the inspection extends 500 m past the LGA and zone boundaries.

Known complexities

- **Scale and distance.** About 3,000 km of road, 1,890 km of it unsealed, across a large rural LGA. JCE's capture vehicle records at traffic speed so that the field program fits between February and April 2027, ahead of the 30 June valuation.
- **Unsealed roads and weather.** Wet weather can close or damage unsealed roads and degrade image quality. The program holds two weeks of contingency and a second hired rig if capture falls behind by mid-March.
- **Matching Council's register.** Ratings, segments and images must link to Council's existing Asset IDs and MapInfo layers. The data review and the Moama pilot settle segment rules, defect codes and naming before full capture.
- **Bridges.** Council requires inspectors with at least five years of bridge condition assessment on concrete, steel and timber bridges, inspection within 3 m of every component, and an access methodology for each structure. Level 3/4 recommendations must reach Council early enough to act on.
- **Valuation timing.** A valuation as at 30/06/2027 that passes audit needs condition data and unit rates settled by May. The valuer reviews the methodology in January and the unit rates in March.
- **Explaining change.** Council asks for every change to its recorded remaining lives or data to be explained. JCE records the reason for each change in the register.

2.0 Methodology and technical approach

JCE will deliver Year 1 in six stages keyed to Council's payment milestones. No work starts on site until Council releases the hold points in writing. The partners work under JCE's direction and quality control; the valuation is prepared with the valuer and signed by the valuer.

Where Schedule 6 is answered:

Schedule 6 item	Section of this proposal
1. Dates for completion	4.0 Program (milestone table and Gantt chart)
2. Understanding, complexities, asset classes and rating system	1.0 Project understanding; 2.0 Stages 2 to 4
3. Inspection method and use of technology	2.0 Stage 3
4. Timeframe and Gantt chart with contingency	4.0 Program, Figure 1
5. QA/QC process	2.0 Quality assurance and quality control
6. Data management, reporting and integration	2.0 Data management, reporting and integration; 3.0 Deliverables
7. Key risks and mitigation	2.0 Key risks and mitigation

Who	Role on MRC-2645
JCE (Jayden Callen, with casual field and GIS support)	Lead Consultant and single point of contact; methodology, hold points, rating guide, Council liaison; vehicle-mounted road imaging and drone capture; condition rating of sealed and unsealed roads, kerb, footpaths, culverts, guardrail, islands, carparks and bus shelters; GIS layers and asset register; remaining life, defect report, guardrail and handrail safety assessment, 10-year programs, condition report, valuation inputs and Director review of every deliverable
Bridge inspection partner	Level 2 inspections and reports for 113 bridges and major culverts and 14 footbridges; access methodology and site safety plans; Level 3/4 inspections if Council takes up the option
Valuer	Valuation methodology, benchmark unit rates, valuation calculations, fair value hierarchy, and the signed valuation report each year
Council	Data at commencement, hold point release, review of the pilot and drafts within the periods in Section 4.0

Stage 1: Mobilisation, Council data review and hold points (JCE)

- Hold point submissions for Council's written release: the written methodology (including plant and outsourced specialists), program with output milestones, list of providers and equipment, methodology for specialist activities, and SWMS and site-specific safety plans.
- Inception meeting and Council WHS induction at Moama, with JCE's agenda and minutes.
- Review of Council's asset register, MapInfo layers, current valuation, as-built dates and WAE plans; set-up of segments (urban block to block with the main road owning the intersection, rural 2 km from zero chainage); mapping of Council's defect schema and Maintenance Management System codes into the rating guide.

Stage 2: Pilot and calibration (JCE, with partners and Council)

- Three-day pilot in Moama covering sealed, unsealed and urban samples, a footpath run, bridges and culverts, so that ratings, image quality, naming, GPS offsets and the data template are confirmed with Council before full capture.
- Calibration of every rater against JCE's rating guide and Council's 1 to 5 scale, with photographic examples of each rating for each component, then Council approval to proceed.

Stage 3: Network inspections (partners and JCE)

- Roads, kerb and roadside (JCE): vehicle-mounted geo-referenced high-resolution stills every 10 m, GPS embedded natively in each image (DRFA compliant), wide enough to capture kerb to kerb or table drain to table drain and roadside hazards on both sides, with offset GPS pick-up of on-road and roadside features. One rig captures the network in three blocks (08/02 to 12/02, 01/03 to 05/03 and 22/03 to 08/04/2027), about 22 capture days at 150 to 200 km a day inside the February to April window.

- Kerb and gutter, footpaths, carparks, roundabouts, islands, medians, bus shelters and rural culverts (JCE): inventory verification, dimensions, condition and defects rated from JCE's vehicle and drone imagery, with at least two images per defect (segment overview and close-up).
- Guardrail and handrail (JCE): inventory, condition and safety data from JCE's vehicle and drone imagery, with unprotected edge drops recorded as defects.
- **Bridges, major culverts and footbridges:** Level 2 visual inspection within 3 m of every component in good natural light (no underwater inspection), component ratings, defects, photographs and an access methodology for each structure.
- Drone capture (JCE): two campaigns, 16 field days in all. Campaign 1 (15/02 to 25/02/2027) runs with the bridge partner and captures bridges, approaches and footbridges where the partner needs overview or hard-to-reach imagery (about 40 structures). Campaign 2 (08/03 to 18/03/2027) captures about 200 rural culverts not visible in the vehicle imagery, stormwater outlets, kerb and footpath hotspots, carparks and assets with poor vehicle access.
- Condition 5 and safety defects notified to Council the same day; weekly progress data.

Vehicle-mounted road imaging (JCE)

- **Equipment.** JCE will deploy a vehicle-mounted 360 or multi-camera system with integrated GNSS and IMU and RTK correction (Mosaic-class or equivalent), recording geotagged stills at 10 m intervals with GPS embedded in each image.
- **Crew.** One rig is enough: about 3,000 km at 150 to 200 km a day is 18 to 22 capture days inside about 60 working days. A casual driver and field assistant drives the capture vehicle; the rig runs automatically. Jayden rides along on the first capture days, checks each day's upload remotely and sets the route plan. Lone and remote work check-ins apply on rural roads.
- **Coverage.** One pass per road, with both sides captured kerb to kerb or table drain to table drain. On single-lane unsealed roads the vehicle drives the centre of the formation so that both table drains are in frame; divided roads are driven in both directions. Routes are planned to limit repositioning to about 4,800 km over the network.
- **Pilot.** The three-day pilot in Moama (27/01 to 29/01/2027) captures sealed, unsealed and urban samples so that Council confirms image quality, naming, GPS offsets, ratings and the data template before full capture.
- **Rating.** Desktop condition rating of sealed roads (surface, pavement, formation), unsealed roads (wearing course, formation), kerb, footpaths and roadside assets against Council's rating guide and 1 to 5 scale, from about 300,000 images. AI-assisted pre-rating may be used to flag likely defects for review; every rating is made or confirmed by a person. The casual technician rates first, and Jayden checks a blind 2% sample in each zone and asset class.
- **Outputs.** Ratings and defects in Council's data template, images named Asset ID, chainage and direction, an image index and the GNSS track.

Drone (RPAS) capture and compliance

- **Equipment.** JCE will deploy a DJI Mavic 3 Enterprise with RTK module (under 2 kg), with a mapping-grade camera for orthophotos, 3D models and geotagged inspection imagery. Workflow: automated flight paths planned in DroneDeploy (or DJI Pilot 2 or DJI FlightHub); RTK corrections from AUSCORS and CORSnet-NSW, or a local base station; ground control and independent check points at each mapped site to verify accuracy; processing in Autodesk ReCap Pro for point clouds and surfaces, with orthophotos from ReCap Pro or DJI Terra; Autodesk InfraWorks and Civil 3D for design integration; QGIS for GIS and asset register delivery.
- **Regulation.** JCE operates under CASA's sub-2 kg excluded category, with operator accreditation and drone registration in place before services start. Every flight is within visual line of sight, below 120 m above ground level, in daylight, at least 30 m from people and never over people.
- **Airspace.** JCE's desk check found no controlled aerodrome within 5.5 km of the LGA; the nearby aerodromes (including Echuca, Swan Hill, Kerang and Deniliquin) are non-controlled. Near a non-controlled aerodrome JCE keeps clear of approach and departure paths and gives way to manned aircraft. JCE confirms airspace with a CASA-verified drone safety app before each flight. [TBC - Jayden: confirm the desk check, including hospital helicopter landing sites near Moama and Echuca]
- **Site controls.** A site risk assessment, pre-flight checklist and flight log for every flight; a casual visual observer on road sites; take-off and landing off the carriageway; traffic and public managed by stopping flights when people or vehicles come within 30 m. No traffic control is used.

- **Existing condition record.** At bridges and approaches captured in campaign 1, JCE will produce RTK-referenced orthophotos and 3D models within the same flight limits, giving Council a baseline of existing condition that the rolling inspections in 2028 to 2031 are compared against.
- **Capability.** Jayden has delivered drone photogrammetry on previous projects, planning automated flight paths in DroneDeploy, capturing RTK-positioned photogrammetry and high-resolution aerial stills and video, and processing in Autodesk ReCap Pro to build point clouds and terrain surfaces for use in Autodesk InfraWorks and civil design models.

Stage 4: Data, remaining life, defects, safety and programs (JCE)

- QA of JCE's road and roadside ratings against Council's template: Asset IDs, segments, components, completeness, image naming and GPS, with a blind re-rated sample in each zone. JCE builds one Excel register and one set of GIS layers in QGIS from its vehicle and drone imagery and ratings, exported to Council's formats.
- Remaining useful life by component and segment from material, age, observed condition, environmental exposure and historical performance, with the reason recorded for every change to Council's data.
- Detailed defect report with repair requirements and program timeframes; guardrail and handrail safety assessment against Austroads Guide to Road Design Part 6 and current Australian Standards, prioritised by risk and severity; review of the bridge reports and a register of recommended Level 3/4 inspections.
- 10-year capital renewal and maintenance programs by asset category, with assumed budget, prioritised works and risk ranking.

Stage 5: Comprehensive valuation as at 30/06/2027 (valuer, with JCE)

- **Valuation methodology:** fair value basis, componentisation (Appendix 2), capitalisation thresholds (Appendix 3) and the fair value hierarchy, reviewed by the valuer before field work ends.
- Current replacement cost unit rates built from Council's contract rates and JCE's estimate data, checked against the valuer's benchmarks; useful lives, residual values and condition-based remaining lives.
- Valuation of every asset above threshold, reconciled by JCE to the register, with working papers; draft and final valuation report signed by the valuer.

Stage 6: Project management, quality and close-out (JCE)

- Single point of contact, monthly progress meetings and reports, monthly consultant certificates, Director review of every deliverable, an online presentation of draft findings, one comments register, and handover with return or deletion of Council's data as the brief requires.

Rolling inspection and valuation, 2028 to 2031

- **Each year:** physical inspection of the nominated zone (Barham 2028, Mathoura 2029, Moama 2030, Moulamein 2031) and of the newly acquired assets Council lists, capture from February to April, an update of the register, defects and programs, and a valuation by 30 June: desktop revaluations of the whole class in 2028 to 2030 and a comprehensive valuation of the whole class in 2031. The same milestones and payment split apply each year.

Quality assurance and quality control

- Rating guide and data template approved by Council after the pilot, with photographic examples of each rating.
- Calibration of every rater before capture, and again if a rater changes.
- Automated completeness checks in QGIS by JCE on every data delivery (Asset IDs, segment continuity, mandatory fields, image naming and GPS).
- Blind re-rate by Jayden of a random sample of at least 2% of the technician's road and non-road ratings in each zone and asset class; a rater whose sample differs by more than one rating point on more than 10% of items re-rates that batch.
- Bridge reports checked by JCE against the Level 2 scope and Council's rating scale before issue.
- Director review of every deliverable before it reaches Council, and independent technical review by Sunny Lotay (RPEQ, NER) of the condition, defect and safety reports and the 10-year programs, recorded on a checklist; one register of Council comments and how each was closed.

Data management, reporting and integration

- **Structure:** Council's Excel data template (tabs Roads, Kerb, Footpath, Bridges, Roundabouts, Culverts, Guardrail, Bus Shelter, Carparks), one row per asset or segment, linked to Council Asset IDs; new assets flagged for Council to assign IDs.
- **Spatial:** GIS layers of segments and point assets built by JCE in QGIS and exported to MapInfo .tab (or Council's preferred format) with Asset ID and inspection metadata; drone orthophotos where captured; coordinates in GDA94 and GDA2020 (MGA zones 54 and 55), heights to AHD.
- **Images:** .jpg with native GPS, named Asset ID, component or chainage, and lane direction or facing, with an image index file.
- **Drone imagery:** JCE's high-resolution aerial stills and short video of the bridges, culverts and hotspots it flies, named by Asset ID and listed in the image index, as existing-condition and defect records Council can also use in reports and presentations.
- **Reports:** preliminary registers and reports in Word and Excel for review; final reports in Word and PDF.
- **Ownership and handover:** all data and images are Council's property, stored in Australia during the work, handed over in full at each milestone, and returned or deleted at completion unless Council consents in writing.

Key risks and mitigation

Risk	Mitigation
Wet weather closes unsealed roads or delays capture	Capture starts 08/02/2027; zones sequenced by forecast; two weeks of contingency to 30/04/2027; short-term hire of a second rig if capture is more than one week behind by mid-March
Council does not accept the image standard	The Moama pilot confirms image quality, naming, GPS offsets and ratings with Council before full capture, and Council signs off the rating guide
Capture rig failure	A spare 360 camera and GNSS receiver (about $5,000 budget) carried on the vehicle, or short-term hire of a second rig, at JCE's cost; the April contingency days absorb lost days
JCE has no previous network road imaging survey	Pilot sign-off by Council, the QA plan with blind re-rating, independent review by Sunny Lotay (RPEQ, NER), and Jayden's drone, RTK and ReCap Pro capture experience
Field data does not match Council's register	Data review in Stage 1 and the pilot in Stage 2 settle Asset IDs, segments and codes before full capture; JCE builds the register itself, so data is checked as it arrives
Drone flights limited by wind, rain or people near the site	Two campaigns with spare days in April; flights stop when people come within 30 m; assets that cannot be flown are rated from JCE's vehicle imagery or photographed from the ground
Airspace or regulatory limits on drone use	Excluded category limits applied to every flight; desk check of aerodromes and airspace confirmed before each flight; no flight over people or within 5.5 km of a controlled aerodrome
Inconsistent ratings between raters	Rating guide, calibration, 2% re-rate sample and batch re-rating rule
Bridge access or safety constraints	Access methodology and site safety plan for each structure as hold points; no under-bridge units, rope access or diving in Level 2
Level 3/4 needs found late	Bridge inspections run from February, ahead of the road reports, and recommendations go to Council as they arise
Valuation not ready for audit	Valuer reviews methodology in January and unit rates in March; draft valuation by 28/05/2027
Lead consultant availability	Bridge inspection is carried by the bridge partner; the casual driver and field assistant and the casual GIS and rating technician carry the capture and rating volume; Sunny Lotay, JCE's associate certifying engineer, is backup project lead; Jayden's time is planned at about 36 to 37 hours a week (Section 5.0)

Sign-off. Reports are prepared and signed by Jayden Callen, BEng (Hons), MIEAust, as JCE's project manager, after independent technical review by Sunny Lotay (RPEQ, NER), JCE's associate certifying engineer, who certifies where a registered engineer is required. Bridge inspection reports are signed by the bridge partner's nominated bridge engineer, and the valuation report by the valuer.

3.0 Deliverables

Deliverable	Format	Target
Hold point submissions: methodology, program and Gantt chart, providers and equipment, SWMS and safety plans	PDF	18/12/2026
Inception agenda and minutes; rating guide and data template confirmed after pilot	PDF, XLSX	12/01/2027; 05/02/2027
Condition 5 and safety defect notices; weekly progress data	Email with images; XLSX	Same day; weekly
Level 2 bridge inspection reports (127 structures) with Level 3/4 recommendations	PDF	25/03/2027 (inspections)
Drone aerial stills, video, orthophotos and 3D models of flown sites (existing-condition and defect records)	JPG, MP4, GeoTIFF, model files	07/05/2027 (draft data)
Preliminary asset register, images and image index, GIS layers (draft data)	XLSX, JPG, GIS	07/05/2027
Draft defect report, guardrail and handrail safety assessment, condition report and 10-year programs	DOCX, XLSX	21/05/2027
Draft valuation report and methodology, with working papers	DOCX, XLSX	28/05/2027
Final asset register, images, GIS mapping (GDA94 and GDA2020, AHD), inspection and condition report, defect report, programs and signed valuation report	XLSX, JPG, GIS, DOCX and PDF	25/06/2027
Rolling zone inspection, register update and valuation report, 2028 to 2031	As above	By 30 June each year

4.0 Program

The brief sets the valuation date of 30/06/2027 and no start date. The program assumes award on Council's expected decision date of 11/12/2026, Council review periods of five business days for hold points and ten business days for draft reports, and no work between 24/12/2026 and 08/01/2027.

Milestone	Target date
Award (Council's expected decision date)	Friday 11/12/2026
Hold point submissions issued	Friday 18/12/2026
Inception meeting and Council WHS induction, Moama	Tuesday 12/01/2027
Council data supplied	Friday 15/01/2027
Hold points released (assumed)	Friday 22/01/2027
Pilot and calibration, Moama	27/01/2027 to 29/01/2027
Pilot approval to proceed	Friday 05/02/2027
JCE vehicle capture, all four zones (one rig, about 22 days)	08/02 to 12/02, 01/03 to 05/03 and 22/03 to 08/04/2027
Capture contingency (if needed)	12/04/2027 to 30/04/2027
Level 2 bridge inspections	15/02/2027 to 25/03/2027
JCE drone campaign 1 (bridges, with the bridge partner)	15/02/2027 to 25/02/2027
JCE drone campaign 2 (culverts, outlets, hotspots)	08/03/2027 to 18/03/2027
Draft data (register, images, GIS)	Friday 07/05/2027
Draft defect, safety and condition reports; 10-year programs	Friday 21/05/2027
Draft valuation report	Friday 28/05/2027
Council comments	By Friday 11/06/2027
Final data and reports; valuation as at 30/06/2027	Friday 25/06/2027
Rolling cycles: zone inspections February to April; reports by 30 June	2028, 2029, 2030, 2031

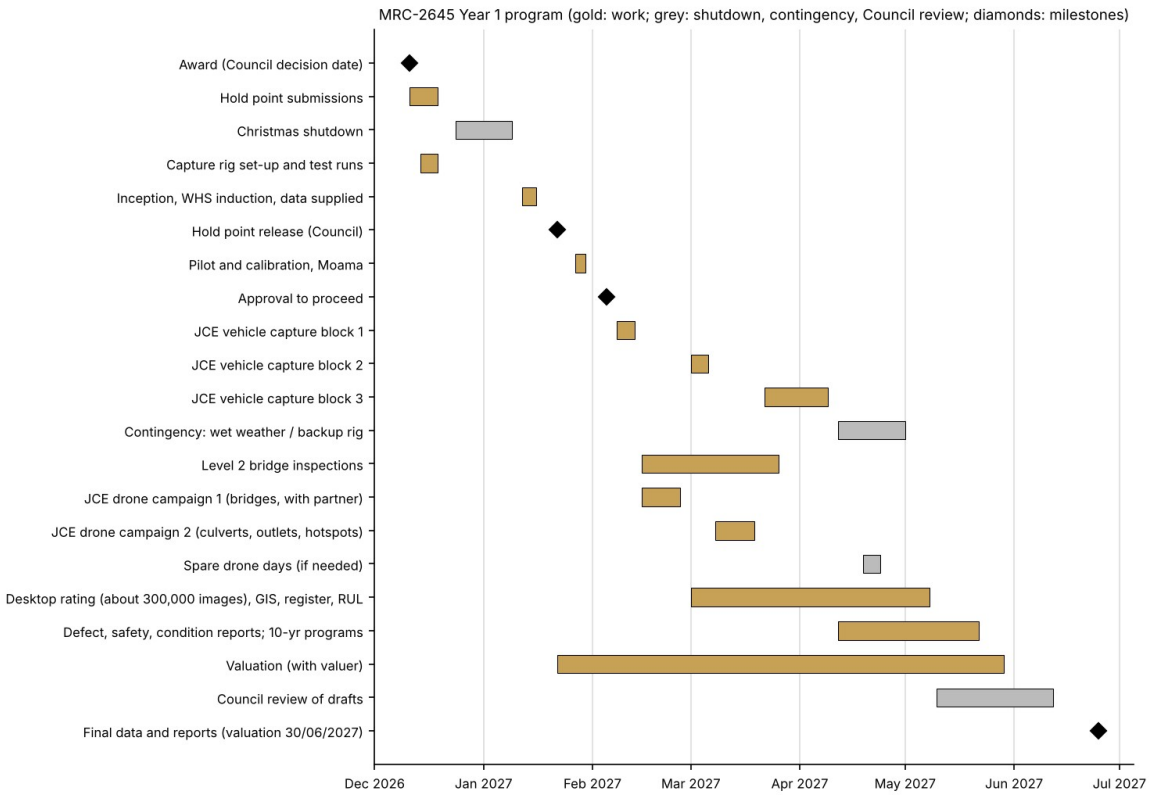


Figure 1: Year 1 Gantt chart, showing contingency and Council review periods.

5.0 Project team, resourcing and experience

Role	Person	Responsibility
Project manager, drone pilot and GIS lead	Jayden Callen, BEng (Hons), MIEAust (JCE)	Single point of contact; methodology, vehicle capture supervision, drone capture, GIS and register, QA, remaining life, defect and safety assessment, programs, valuation inputs and reports
Associate Principal Engineer, technical reviewer and certifying engineer	Sunny Lotay, Associate Principal Engineer (RPEQ, NER), Lotay Civil	Engaged by JCE under a standing associate agreement as technical reviewer and certifying engineer on JCE commissions: independent technical review of the condition, remaining life, defect and safety reports and 10-year programs; certification where a registered engineer is required; backup project lead
Casual driver, field assistant and visual observer	[TBC - Jayden: name]	Drives the capture vehicle (about 22 capture days and the pilot), drone visual observer on road sites, data offload and site safety
Casual GIS and rating technician	[TBC - Jayden: name]	First-pass rating of road and non-road assets from about 300,000 images, GIS layers, register population, image index and maps, checked by Jayden
Bridge inspection partner	[TBC - Jayden: firm selected after quotes due 28/10/2026]	Level 2 bridge, major culvert and footbridge inspections and reports; Level 3/4 if Council takes up the option
Valuer	[TBC - Jayden: firm selected after quotes due 28/10/2026]	Valuation methodology, unit rate benchmarks, calculations and signed valuation report

How JCE's own hours are delivered

JCE is a principal-led practice. Jayden Callen delivers JCE's own work: about 923 hours between December 2026 and June 2027, an average of about 36.4 hours a week over about 25 working weeks.

Two casual staff carry the volume work under his direction: a driver and field assistant for about 22 capture days, the pilot and the drone days, and a GIS and rating technician for first-pass ratings of about 300,000 images, GIS layers, the register and report tables (about 720 hours). The bridge partner inspects the structures and the valuer runs the valuation calculations.

Period	Main JCE activities	JCE hours (approx.)
Dec 2026	Hold point submissions, capture rig set-up and test runs, drone operations plan, GIS set-up	74
Jan 2027	Inception, data review, segmentation, pilot with rig and drone	112
Feb 2027	Capture block 1 with ride-along, drone campaign 1 with the bridge partner, daily upload QC	149
Mar 2027	Capture block 2, drone campaign 2, bridge report reviews, unit rates with valuer	160
Apr 2027	Capture block 3, road rating QA, remaining life model, register and GIS	146
May 2027	Draft defect, safety and condition reports, 10-year programs, valuation inputs	160
Jun 2027	Council comments, final reports and data, close-out	122
Total		**923**

Weekly average by month: December 37, January 37.3, February 37.3, March 36.4, April 36.5, May 36.4 and June 34 hours. December to April run above 36 hours; the technician takes the completeness checks and report tables, and Jayden takes on no other major commission from December to May.

Drone photogrammetry capability

JCE's related business, Glide Inside (glideinside.com.au), produces 3D digital models and visualisation for the property and construction sector. JCE will deliver drone photogrammetry (orthophotos, 3D models and geotagged inspection imagery) using a DJI Mavic 3 Enterprise with RTK, operated under CASA's sub-2 kg excluded category (operator accreditation and drone registration in place before services start). Jayden has delivered drone photogrammetry on previous projects, planning automated flight paths in DroneDeploy, capturing RTK-positioned photogrammetry and high-resolution aerial stills and video, and processing in Autodesk ReCap Pro to build point clouds and terrain surfaces for use in Autodesk InfraWorks and civil design models.

Experience

Jayden Callen is a civil engineer with more than eight years of experience across Commonwealth Defence infrastructure, land development, roads and pavements, stormwater and water infrastructure. Experience relevant to MRC-2645:

- **Managing specialists to one program.** Civil Design Manager at Langtree Consulting Engineers, Townsville (03/2025 to 05/2026): managed four civil designers, coordinated flood modelling, electrical and hydraulic subconsultants into joint packages including bridge approaches, culverts and retaining walls, and prepared engineer's estimates and bills of quantities. Senior Engineer and Design Manager at SCP Consulting (06/2023 to 03/2025): coordinated external subconsultants and reviewed technical deliverables before client release on Defence projects at HMAS Watson, RAAF Base Tindal, RAAF Base Townsville and HMAS Cairns.
- **Council road network works.** Jayden Callen (while at Langtree Consulting Engineers): Floodway rectification, Tablelands Regional Council (QLD): engineering for the rectification of a floodway on Council's road network. Floodways and causeways are among the transport assets in this tender. Referee: Darren Quilliam, Tablelands Regional Council, contact details available on request.
- **Council client work.** Jayden Callen (while at Langtree Consulting Engineers): Leachate storage facility, Tablelands Regional Council (QLD): engineering for a landfill leachate storage facility. Referee as above.
- **Regional and remote delivery.** Civil design management, subconsultant coordination and site inspections for subdivisions in the remote Cape York communities of Aurukun, Pormpuraaw and Napranum (Langtree).
- **Drone photogrammetry project.** Farm drainage works, private rural client (Craig Sullivan): drone photogrammetry survey of the property using RTK-positioned automated flights, processed in

Autodesk ReCap Pro to a terrain surface used to design the drainage works. Referee details available on request.

- **Defence head-contractor delivery.** RAAF Base Townsville, Defence delivery project: engineering delivery for CPB Contractors (head contractor). Referee: Maverick Partner, CPB Contractors, contact details available on request.
- **Roads and costs.** Road geometry to the Austroads guides with Transport for NSW supplements, pavement design with CIRCLY, and engineer's estimates and bills of quantities, which are the basis of replacement cost unit rates.
- **Data, drone and GIS.** Civil 3D, Navisworks and InfraWorks; CAD standards and automation in AutoLISP and Excel VBA; Autodesk ReCap Pro, DroneDeploy and QGIS (drone capability above).

JCE has not yet delivered a network road imaging survey, condition audit or valuation as lead consultant. The pilot, the QA plan, Sunny Lotay's independent review and Jayden's drone, RTK and ReCap Pro experience are how JCE manages that. The bridge partner's and valuer's staff, comparable council projects and referees are set out in Schedules 3 to 5 and Attachment B: [TBC - Jayden: insert from partner quotes due 28/10/2026].

6.0 Work health and safety, quality and environment

Work health and safety. JCE operates a documented WHS management system: WHS Policy (JCE-WHS-POL-001), WHS Management Plan (PLN-001), Safe Work Method Statement (SWMS-001), Site Inspection and Site Entry Procedure (PRO-005), Lone and Remote Worker Procedure (PRO-006), Driving and Vehicle Safety Policy (POL-004), Fatigue Management Policy (POL-005) and Subconsultant and Contractor WHS Management Procedure (PRO-009). SWMS and site-specific safety plans are hold point submissions. Imaging vehicles travel at traffic speed with warning lights, bridge inspections follow an access methodology for each structure, staff wear AS 4602 high-visibility clothing and AS 2210 footwear, and no road or bridge is closed without Council's agreement. Long regional drives follow JCE's fatigue and journey planning rules.

Drone and road-side safety. Drone flights follow CASA's sub-2 kg excluded category rules (visual line of sight, below 120 m above ground level, daytime, at least 30 m from people, no flight over people, not within 5.5 km of a controlled aerodrome), with a site risk assessment, pre-flight checks, a flight log and a visual observer at road sites. Flights pause when people or vehicles approach within 30 m. JCE's capture vehicle travels at traffic speed with a roof beacon and survey vehicle signage under a vehicle capture SWMS; the driver only drives, the rig runs automatically, and no traffic control or closures are used. CASA rules do not apply to vehicle capture. Drone public liability insurance: [TBC - Jayden: through Webber, target $10 million to $20 million].

Quality system and ISO 9001. Item 12 and clause 3.8 of the Agreement require the Consultant to establish and maintain a quality assurance system that complies with ISO 9001. JCE is not third-party certified to ISO 9001. JCE operates a documented quality system under its Quality Policy (JCE-QMS-POL-001): document and version control, a structured job numbering and filing system, checking and Director review before issue, and management of subconsultant work to the same standard (PRO-009). For MRC-2645 JCE will prepare a project quality plan that follows the ISO 9001 requirements for planning, control of externally provided services, monitoring and measurement, nonconformity and corrective action, and records, submitted with the hold point documents and open to audit by Council's Representative under clause 3.9. JCE will ask each partner for its ISO 9001 status and quality records. [TBC - Jayden: state any ISO 9001 certification plan and timeframe, or delete this note.]

Environment, heritage and privacy. Inspections are made from formed roads and road reserves with no ground disturbance. Vehicles stay on formed surfaces, and work stops and Council is notified on any unexpected cultural heritage find. Imagery and Council data are stored in Australia and returned or deleted at completion unless Council consents in writing (Environmental Policy JCE-ENV-POL-001).

7.0 Local content

JCE is based outside the Murray River Council area and has no staff or depot in the region. Its economic contribution under this contract is local spend during field work: accommodation, meals and fuel in Moama, Barham, Mathoura and Moulamein for JCE's field work (about 64 room-nights in Year 1, vehicle capture and drone campaigns) and for the bridge partner's crews during the February to April 2027 field

program, and the same each year from 2028 to 2031 in the zone being inspected. [TBC - Jayden: add any partner staff, office or suppliers in the region or in Echuca, and a local casual field assistant if one is engaged.]

8.0 Assumptions and information required from Council

Information required from Council

- **Within five business days of award:** the asset register (Excel) with Asset IDs, asset locations in MapInfo .tab, the current valuation and working papers, as-built dates, WAE plans, aerial imagery and photographs where available, and Council's defect schema.
- Council's recent contract rates for road, bridge and footpath renewal and maintenance works, and the list of newly acquired assets each year for the rolling cycles.
- A letter of authority for field staff, notice to residents through Council's channels, and access to a Council depot for induction.

Assumptions

- Award by 11/12/2026 and hold point release within five business days; a later award or release moves the program by the same period.
- Field work in daylight hours, Monday to Friday, from public road reserves without traffic control.
- Drone flights within CASA's excluded category limits; assets that cannot be flown within those limits are rated from JCE's vehicle imagery or ground photographs.
- Level 2 bridge inspections from ground, bank and deck level within 3 m of components, without under-bridge units, rope access, confined space entry or diving.
- One round of consolidated Council comments on each draft report.
- The 2028 to 2031 prices allow for nominal network growth of up to 2% a year and include escalation of 3% a year.

9.0 Fee schedule

Council's Schedule 1 asks for one lump sum inclusive of GST, broken down by valuation year and asset category. The partners' services are shown as provisional sums (PS) at cost plus JCE's 10% mark-up until their quotes are received.

Item	Schedule 1 breakdown by valuation year	Total (excl. GST)
1	Year 1 (2027): comprehensive inspection, defects, safety assessment, 10-year programs and comprehensive valuation as at 30/06/2027	$ 499,300.00
2	2028: Zone 1 Barham inspection and desktop revaluation	$ 152,180.00
3	2029: Zone 2 Mathoura inspection and desktop revaluation	$ 160,420.00
4	2030: Zone 3 Moama inspection and desktop revaluation	$ 138,200.00
5	2031: Zone 4 Moulamein inspection and comprehensive valuation	$ 205,330.00
	Lump sum (excl. GST)	**$ 1,155,430.00**
	GST (10%)	**$ 115,543.00**
	Lump sum for Schedule 1 (incl. GST)	**$ 1,270,973.00**

Part	Breakdown of Item 1 (Year 1)	Amount (excl. GST)
Stage 1	Mobilisation: inception, Council data review, hold point submissions, capture rig set-up and drone operations plan	$ 22,280.00
Stage 2	Pilot, calibration and rig and drone trial	$ 6,600.00
Stage 3	Capture supervision and daily upload QC, drone capture (16 field days), field travel and bridge report review	$ 47,930.00
Stage 4	Road and non-road rating QA, GIS and register, remaining life, defect report, safety assessment, 10-year programs and condition report	$ 59,090.00
Stage 5	Valuation inputs, reconciliation and valuer coordination	$ 13,720.00
Stage 6	Project management, quality review and close-out	$ 21,600.00
AS	Associate certifying engineer (Sunny Lotay, Lotay Civil): independent technical review and certification of JCE reports, 24 h at $220/h allowance (cost plus 10%) [TBC - Jayden to confirm rate]	$ 5,808.00

Part	Breakdown of Item 1 (Year 1)	Amount (excl. GST)
K1	Casual driver and field assistant: about 22 capture days, 3-day pilot, drone visual observer days, data offload and travel (fixed allowance) [TBC - Jayden]	$ 10,000.00
K2	Casual GIS and desktop rating technician: first-pass rating of about 300,000 road and roadside images and drone imagery, road defect coding, image processing, GIS layers, register population, image index, maps and report tables (718 h at $110/h charge rate; cost about $65/h)	$ 78,980.00
EQ1	Drone equipment day rate (DJI Mavic 3 Enterprise with RTK, batteries, RTK corrections, processing software and storage): 17 field days at $250 per day [TBC - Jayden to confirm rate]	$ 4,250.00
EQ2	Vehicle capture rig equipment rate (one 360/multi-camera system with GNSS/IMU and RTK, mounts, storage; covers lease or amortised purchase): 25 capture and pilot days at $350 per day [TBC - Jayden to confirm rate]	$ 8,750.00
EQ3	Rating and image review software allowance, including AI-assisted pre-rating if used [TBC - Jayden]	$ 3,000.00
RK	Rig backup: spare 360 camera and GNSS receiver, or short-term hire of a second rig if the first fails	$ 5,000.00
TR	Vehicle, accommodation and meals for capture, drone campaigns and pilot [TBC - Jayden to confirm rates]	$ 30,352.00
PS2	Bridge inspection partner: Level 2 inspections of 127 structures (provisional sum, cost plus 10%)	$ 129,140.00
PS3	Valuer: methodology and unit rate review, signed valuation report (provisional sum, cost plus 10%)	$ 33,000.00
PS3b	Valuer: valuation calculations and benchmark unit rates (provisional sum, cost plus 10%)	$ 19,800.00
	Item 1 (excl. GST)	**$ 499,300.00**

No	Year 1 by asset category	Total (excl. GST)
1	Sealed roads (urban, regional, local)	$ 131,030.60
2	Unsealed roads	$ 136,342.64
3	Kerb and gutter	$ 6,925.34
4	Footpaths and cycleways	$ 17,457.63
5	Roundabouts, traffic islands and medians	$ 4,898.89
6	Carparks	$ 9,915.83
7	Rural culverts	$ 36,712.17
8	Guardrail	$ 16,395.22
9	Bus shelters	$ 1,377.20
10	Bridges and major culverts	$ 129,063.16
11	Footbridges	$ 9,181.32
	Year 1 (excl. GST)	**$ 499,300.00**

Option (not in the lump sum). Level 3/4 bridge inspection, including component visual rating, structural safety inspection, load capacity assessment, component inventory and photographic record of condition 4 and 5 defects: $ 7,400.00 excl. GST ($ 8,140.00 incl. GST) per bridge.

The lump sum includes all travel and expenses. Additional work directed by Council is valued at the rates in Attachment A.

10.0 Payment terms

JCE will claim on the last day of each month against Council's milestones for each year: mobilisation 20%, inspections complete 30%, draft data 25%, draft report 15%, and final data and report 10%, as Item 6 of the Consultancy Agreement provides.

11.0 Exclusions and variations

The following are excluded unless Council directs them as a variation:

- Level 3/4 bridge inspections and load rating (priced as the option), underwater inspection, and under-bridge, rope access or confined space access.
- Traffic control and road or bridge closures.
- Design and documentation of repairs, renewals or upgrades, and delivery of works.
- Assets owned by others (including State roads and Transport for NSW bridges), stormwater pits and pipes other than road culverts, and valuation of street furniture and signage (inspected and recorded only, as Appendix 3 provides).
- Entry of data into Council's corporate systems (JCE supplies import-ready files in Council's template).

Variations. Changes directed by Council under the Consultancy Agreement are valued at the rates in Attachment A, with a quote first where Council asks for one.

12.0 Insurance

Item 18 and clause 14.1 of the Consultancy Agreement require public liability insurance of $20 million for any one occurrence, professional indemnity insurance of $5 million maintained for seven years after Completion, professional indemnity of $5 million for each subconsultant, and workers compensation.

Policy	JCE cover (certificate of currency issued 13/07/2026)	Position against Item 18
Professional indemnity	$10 million any one claim, $20 million aggregate; Certain Underwriters at Lloyd's led by Markel (Syndicate 3000); excess $1,000; policy 05-ENG-I-0077656 to 02/07/2027	Meets $5 million
Public liability	$10 million; Zurich Australian Insurance Limited; excess $500; same policy	JCE will lift public liability to $20 million on award, before the Agreement is signed, and provide the certificate of currency
Workers compensation	[TBC - Jayden: icare policy number; needed for the casual staff]	Held as required by law
Vehicle and capture rig	JCE business motor policy plus portable equipment / general property cover for the rig and vehicle during capture [TBC - Jayden: being arranged through Webber]	In place before capture starts
Drone (aviation) liability	[TBC - Jayden: through Webber; target $10 million to $20 million. Check whether the broadform policy excludes aircraft]	To be in place before drone flights start
Subconsultants	Each partner to hold professional indemnity of at least $5 million for seven years, public liability and workers compensation	Required in each subconsultancy

13.0 Contract, conflict of interest and validity

Contract. JCE has reviewed Council's draft Consultancy Services Agreement and offers to contract on it. JCE will seek Council's written approval of each subconsultant under clause 8 and will pass down the Agreement's insurance, WHS, confidentiality and records obligations.

Conflict of interest. JCE has no actual or potential conflict of interest with Council or this tender. Each partner will declare any previous work for Council on these assets and any relationship with Council staff before JCE names it.

Validity. This offer remains open for acceptance for sixty (60) days from the closing time, as clause 3.8 of the Conditions of Tender requires.

14.0 Offer and acceptance

JCE offers to provide the services described in this proposal for the prices in Section 9.0, on the basis set out in Sections 8.0 and 13.0.

Offered by	**Accepted for and on behalf of Murray River Council**
[signature] Jayden Callen, BEng (Hons), MIEAust Director, JCE Infrastructure Pty Ltd ABN 29 699 165 663 Date: ____ / ____ / 2026	Name: _____ Position: _____ Signature: _____ Date: ____ / ____ / _____

Attachment A: Schedule of rates

Applies to variations directed by Council. Rates are exclusive of GST; Schedule 1 shows them including GST.

JCE role	Rate (excl. GST)	Rate (incl. GST)
Director / Principal Engineer	$200/hr · $1,600/day	$220.00/hr
Senior Engineer	$190/hr · $1,520/day	$209.00/hr
Project Engineer	$180/hr · $1,440/day	$198.00/hr
CAD / GIS Technician	$160/hr · $1,280/day	$176.00/hr
Administration / Document Control	$140/hr · $1,120/day	$154.00/hr
Vehicle travel	$0.96/km	$1.06/km
Drone equipment (with JCE hourly rates for flying and processing)	$250/field day	$275.00/field day
Partner services	At cost + 10%	At cost + 10%

Additional JCE resources for directed work: vehicle capture rig $350 per capture day plus a driver at cost plus 10%; GIS and rating technician $110 per hour. Partner rates for directed additional work, including JCE's 10% mark-up: bridge inspector per hour and valuer per hour [TBC - Jayden: from partner quotes, parts B and C]. A day comprises eight (8) hours. Disbursements other than vehicle travel are charged at cost.

Attachment B: Key personnel

Jayden Callen, BEng (Hons), MIEAust. Director and Principal Engineer, JCE Infrastructure. Project manager, drone pilot and GIS lead, and single point of contact. Bachelor of Engineering (Honours) Civil, Deakin University; Engineers Australia member 11036201; application for the National Engineering Register (civil) in progress. Drone operations under CASA's sub-2 kg excluded category (operator accreditation and drone registration in place before services start). Jayden has delivered drone photogrammetry on previous projects, planning automated flight paths in DroneDeploy, capturing RTK-positioned photogrammetry and high-resolution aerial stills and video, and processing in Autodesk ReCap Pro to build point clouds and terrain surfaces for use in Autodesk InfraWorks and civil design models. Detailed CV attached.

Sunny Lotay, Associate Principal Engineer (RPEQ, NER), Lotay Civil: engaged by JCE under a standing associate agreement as technical reviewer and certifying engineer on JCE commissions. His role on MRC-2645 is independent technical review of JCE's outputs, certification where a registered engineer is required, and backup project lead.

Partner key personnel: names, qualifications, years of experience and comparable council projects for the bridge inspection partner (at least five years of bridge condition assessment of concrete, steel and timber bridges; TfNSW experience desirable) and the valuer. [TBC - Jayden: from partner quotes due 28/10/2026]